

SAN LUIS, Argentina

WMO: 874360

Lat: 33.2744S

Lon: 66.3514W

Elev: 713

StdP: 93.05

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	0.4	2.0	-12.4	1.4	11.0	-9.9	1.8	10.9	14.7	13.6	13.0	13.3	2.6	50	0.506

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.6	35.0	20.6	33.6	20.1	32.2	19.7	23.0	31.0	22.2	30.3	21.5	29.4	5.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.6	16.7	27.6	19.7	15.7	26.4	18.9	14.9	25.8	72.5	31.0	69.2	30.5	66.1	29.5	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.8	11.9	10.4	-2.7	37.8	2.1	1.5	-4.2	38.9	-5.5	39.8	-6.6	40.6	-8.1	41.7
			-4.5	24.9	1.5	1.2	-5.5	25.7	-6.4	26.4	-7.2	27.1	-8.3	27.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.1	25.0	23.6	21.3	17.3	13.9	11.2	10.8	13.0	15.8	19.1	21.8	24.1	(6)
	DBStd	6.18	3.02	3.46	3.40	3.66	3.63	3.30	3.86	4.39	4.48	4.16	3.91	3.50	(7)
	HDD10.0	99	0	0	0	2	8	25	36	19	7	1	0	0	(8)
	HDD18.3	994	1	4	13	59	142	213	234	173	98	41	13	3	(9)
	CDD10.0	3041	465	381	350	222	129	62	61	113	182	283	356	438	(10)
	CDD18.3	895	207	151	104	29	5	0	1	8	23	64	118	183	(11)
	CDH23.3	8773	2168	1442	838	219	33	3	13	106	236	607	1177	1932	(12)
	CDH26.7	3468	974	594	266	36	2	0	2	28	61	194	455	856	(13)
(14) Wind	WSAvg	4.6	5.0	4.6	4.3	3.8	3.5	3.6	3.9	4.7	5.1	5.7	5.5	5.2	(14)
(15) Precipitation	PrecAvg	629	109	100	91	41	16	8	6	9	20	43	80	101	(15)
	PrecMax	986	213	368	203	148	107	75	27	55	94	126	189	228	(16)
	PrecMin	230	11	29	3	0	0	0	0	0	0	0	10	1	(17)
	PrecStd	149	50	60	52	37	22	13	6	13	22	32	47	51	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.7	35.7	33.2	29.9	26.2	23.5	24.9	29.8	31.4	33.5	34.7	36.8	(19)
		MCWB	21.1	21.3	21.0	19.6	16.7	13.7	14.7	15.5	16.5	18.1	19.5	21.4	(20)
	2%	DB	34.9	33.5	31.2	27.6	24.0	21.1	21.9	26.3	28.3	30.8	32.9	34.8	(21)
		MCWB	21.1	20.9	20.1	18.1	16.0	12.6	12.3	13.6	15.0	17.1	18.6	20.4	(22)
	5%	DB	33.2	32.0	29.7	25.9	22.3	19.2	19.8	23.3	26.0	28.8	31.1	33.0	(23)
		MCWB	20.5	20.8	19.6	17.0	14.8	11.6	11.1	12.5	13.9	16.0	18.0	19.8	(24)
	10%	DB	31.6	30.5	28.1	24.1	20.4	17.5	17.7	20.8	23.9	26.8	29.4	31.2	(25)
		MCWB	20.1	20.1	19.0	16.3	14.0	10.7	9.9	11.2	12.9	15.2	17.4	19.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.3	24.1	23.1	21.5	18.4	16.0	16.0	17.1	17.9	20.4	21.8	23.5	(27)
		MCDB	32.5	31.1	30.4	27.3	23.5	20.9	22.7	26.8	28.8	29.8	31.5	32.3	(28)
	2%	WB	22.9	23.0	21.8	19.8	17.1	14.0	13.9	15.1	16.4	18.6	20.3	22.2	(29)
		MCDB	31.4	30.4	28.9	24.4	22.1	19.0	19.5	23.7	25.7	27.1	29.2	31.3	(30)
	5%	WB	22.0	22.1	20.7	18.6	16.1	12.7	12.3	13.4	14.9	17.4	19.2	21.3	(31)
		MCDB	30.4	29.4	27.1	23.4	21.1	17.5	17.6	21.3	23.8	25.9	28.7	30.2	(32)
	10%	WB	21.1	21.1	19.7	17.5	14.8	11.5	10.8	11.9	13.7	16.4	18.3	20.3	(33)
		MCDB	29.3	28.4	26.1	22.7	19.2	16.2	16.5	19.8	21.9	24.6	27.4	28.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	11.6	11.2	10.5	10.2	9.9	10.7	11.4	11.8	11.7	11.4	11.8	11.9	(35)
		MCDBR	13.2	12.9	12.5	12.8	12.2	13.1	14.0	14.9	14.0	13.7	13.4	13.6	(36)
	5% WB	MCWBR	5.3	5.3	5.6	6.1	6.2	7.0	7.2	7.2	6.6	6.2	5.8	5.9	(37)
		MCWBR	11.5	11.2	10.9	10.3	10.4	11.3	12.4	13.4	12.5	12.1	12.3	12.2	(38)
(40) Clear-Sky Solar Irradiance	taub		0.372	0.366	0.352	0.343	0.315	0.302	0.303	0.332	0.376	0.365	0.363	0.370	(40)
		taud	2.430	2.449	2.485	2.479	2.509	2.548	2.501	2.405	2.275	2.363	2.398	2.409	(41)
	Ebn at Noon		966	951	929	880	855	845	860	877	887	941	968	971	(42)
		Edh at Noon	122	116	104	94	80	72	79	99	126	125	126	126	(43)
(44) All-Sky Solar Radiation	RadAvg		7.64	6.65	5.41	4.00	2.94	2.75	3.02	3.96	5.14	6.30	7.41	7.89	(44)
	RadStd		0.40	0.44	0.46	0.42	0.34	0.23	0.20	0.29	0.42	0.54	0.42	0.39	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	+0.57	N/A	+0.43	+0.81	+0.82	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	N/A	+0.56	+0.62	+0.78	N/A	N/A	N/A

Nomenclature: See separate page